



**Sunbeam Institute of Information Technology**  
**Pune and Karad**

## **Module - Operating System Concepts and Linux Programming**

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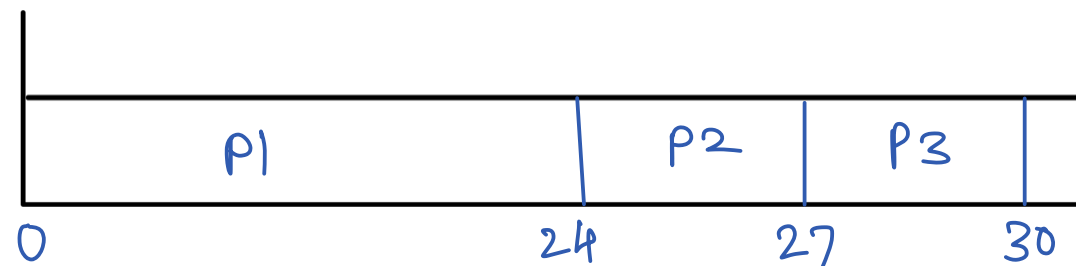
# FCFS (First Come First Serve) (Non preemptive)

| Process | Arrival | CPU Burst |
|---------|---------|-----------|
| P1      | 0       | 24        |
| P2      | 0       | 3         |
| P3      | 0       | 3         |

RT WT TAT  
 0 0 24  
 24 24 27  
 27 27 30

| Process | Arrival | CPU Burst |
|---------|---------|-----------|
| P3      | 0       | 3         |
| P2      | 0       | 3         |
| P1      | 0       | 24        |

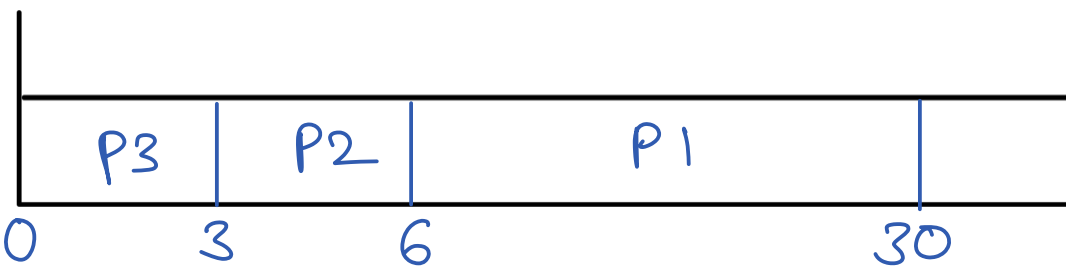
RT WT TAT  
 0 0 3  
 3 3 6  
 6 6 30



0  
 P1  
 P2  
 P3

Convoy effect:

- if longer process is arrived early all other processes need to wait for longer duration.



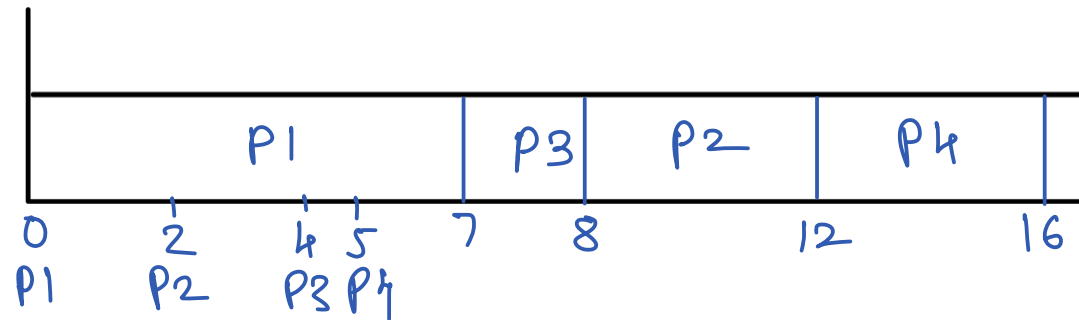
0  
 P3  
 P2  
 P1

# SJF (Shortest Job First)

(Non pre emptive)

| Process | Arrival | CPU Burst |
|---------|---------|-----------|
| P1      | 0       | 7         |
| P2      | 2       | 4         |
| P3      | 4       | 1         |
| P4      | 5       | 4         |

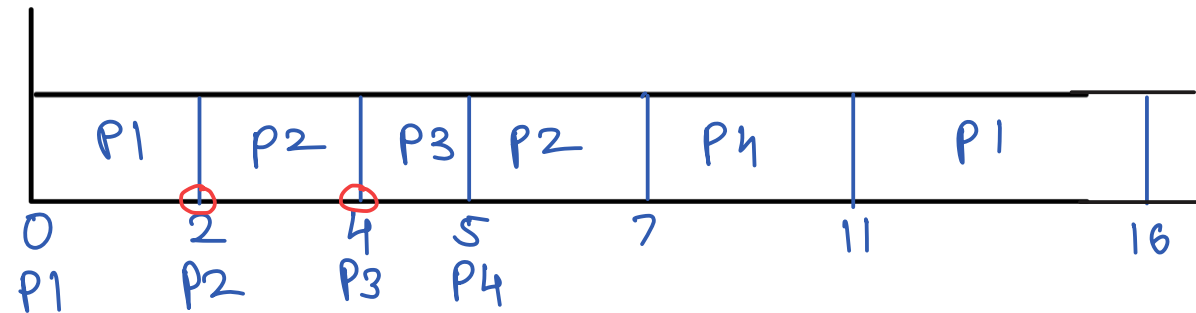
| RT | WT | TAT |
|----|----|-----|
| 0  | 0  | 7   |
| 6  | 6  | 10  |
| 3  | 3  | 4   |
| 7  | 7  | 11  |



Shortest Remaining Time first  
(pre emptive)

| Process | Arrival | CPU Burst |
|---------|---------|-----------|
| P1      | 0       | 7         |
| P2      | 2       | 4         |
| P3      | 4       | 1         |
| P4      | 5       | 4         |

5  
2X  
X  
X



Starvation:

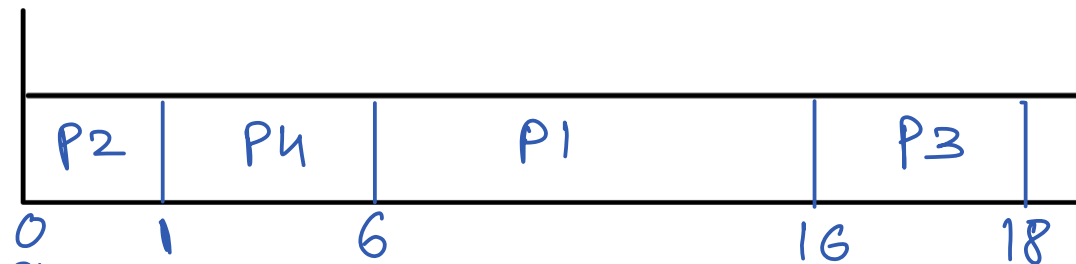
due to longer CPU burst, process need to wait for longer duration in ready queue to get CPU access.

# Priority

Non pre emptive

| Process | Arrival | CPU Burst | Priority |
|---------|---------|-----------|----------|
| P1      | 0       | 10        | 3        |
| P2      | 0       | 1         | 1 (H)    |
| P3      | 0       | 2         | 4 (L)    |
| P4      | 0       | 5         | 2        |

RT WT TAT  
 6 6 16  
 0 0 1  
 16 16 18  
 1 1 6



0  
P1  
P2  
P3  
P4

P1 (6)  
 P2 (8)  
 P3 (2)  
 P4 (5)  
 P5 (6)  
 P6 (6)  
 P7 (6)

P1  
 P4  
 P5  
 P3  
 P6  
 P7  
 P2

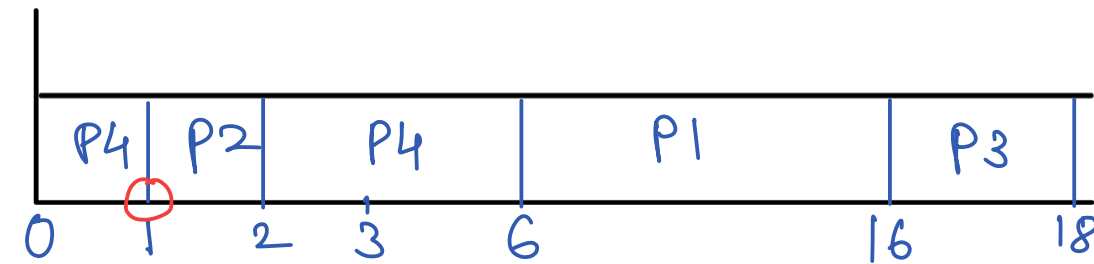
Starvation:

due to low priority process need to spend more time in ready queue to get CPU access

pre emptive

| Process | Arrival | CPU Burst | Priority |
|---------|---------|-----------|----------|
| P1      | 0       | 10        | 3        |
| P2      | 1       | 1         | 1 (H)    |
| P3      | 3       | 2         | 4 (L)    |
| P4      | 0       | 5         | 2        |

RT WT TAT  
 6 6 16  
 0 0 1  
 13 13 15  
 0 1 6



0  
P1  
P4

1  
P2

3  
P3

Aging:

increasing the priority of a process gradually.

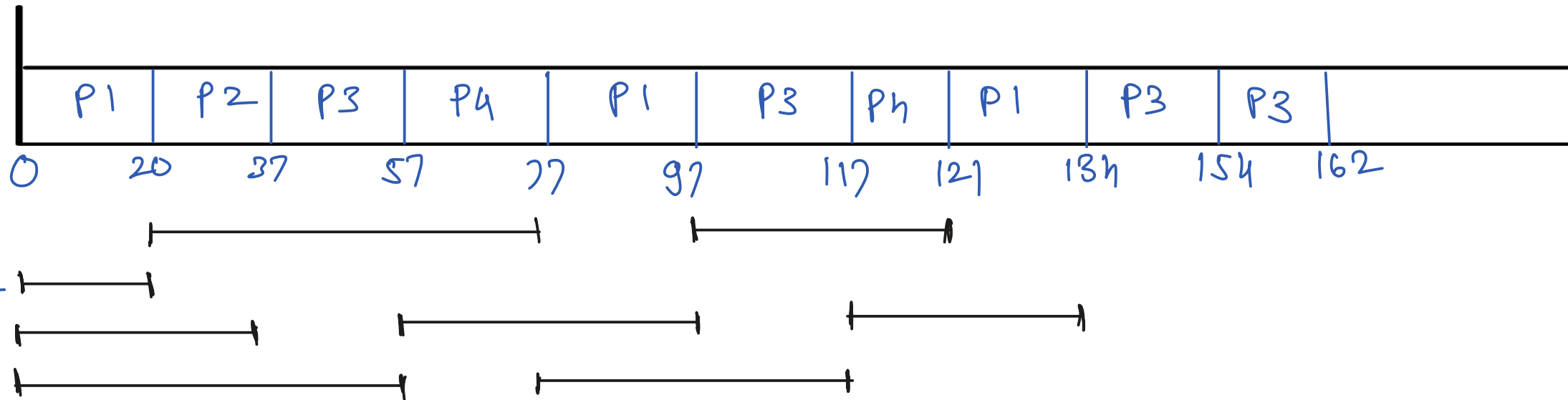
# RR (Round Robin)

$TQ = 20$

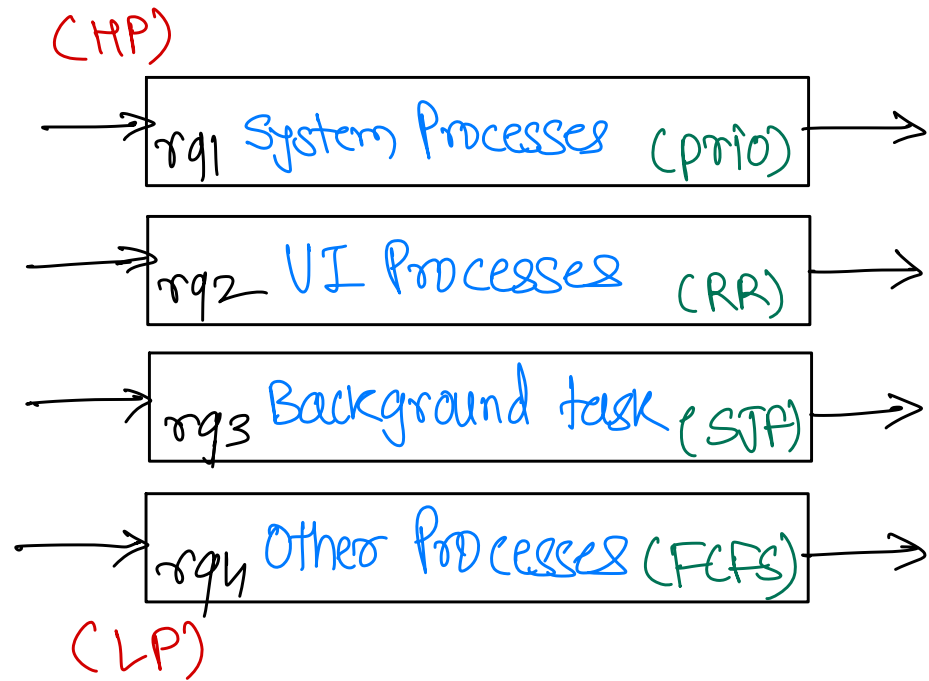
| Process | CPU Burst |
|---------|-----------|
| P1      | 53        |
| P2      | 17        |
| P3      | 68        |
| P4      | 24        |

|           |              |    |
|-----------|--------------|----|
|           | WT           | RT |
| 33, 13    | 0 + 57 + 24  | 0  |
|           | 20           | 20 |
| 48, 28, 8 | 37 + 40 + 17 | 37 |
| 4         | 57 + 40      | 57 |

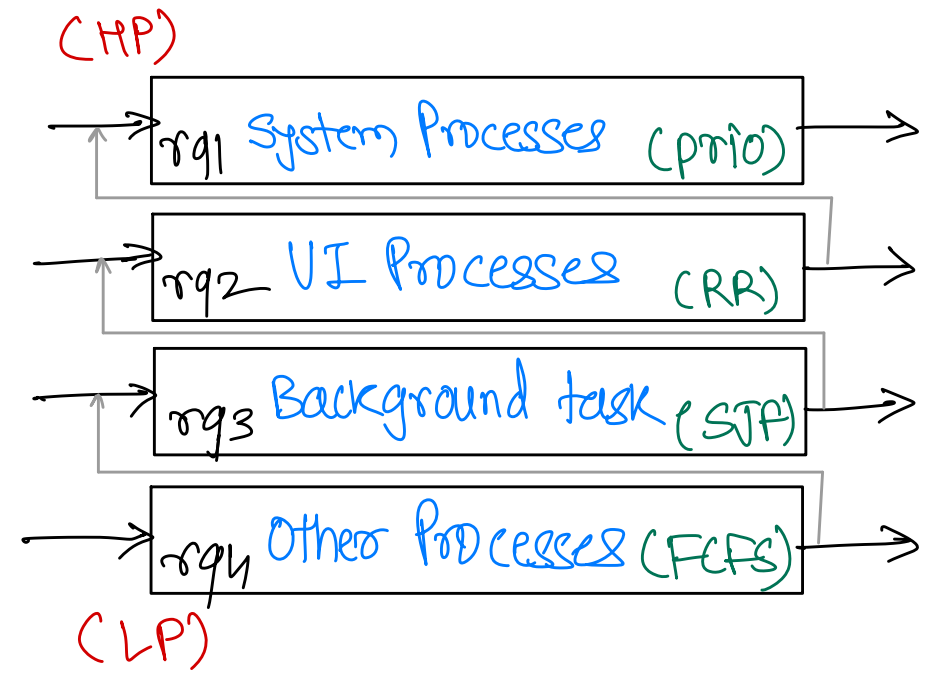
- CPU time is divided into multiple slices which is called as time quantum
- in every time quantum next process in ready queue is scheduled.



# Multi Level Ready Queue



Multi level



Multi Level Feedback

## Linux scheduling policies

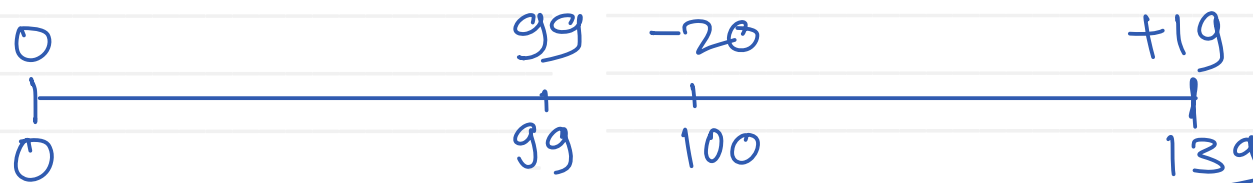
1. Real time  
(high priority process never waits for low priority)

2. Non real time  
(high priority process may need to wait for low process)

## Process classes in Linux

1. SCHED\_FIFO (preemptive FFS) 0 - 99  
2. SCHED\_RR (RR)

3. SCHED\_OTHER  
4. SCHED\_BATCH  
5. SCHED\_IDLE } (CFS) -20 to +19



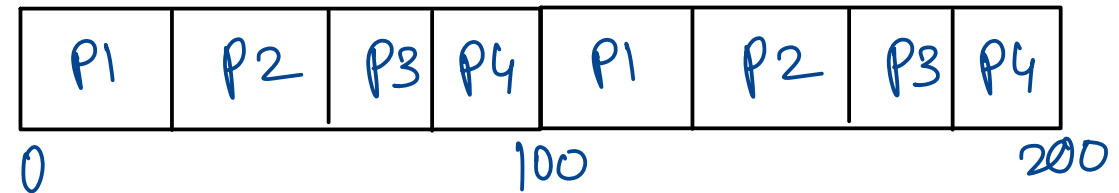
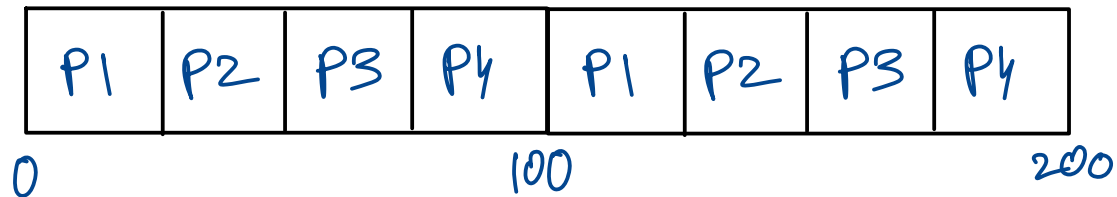
# Fair Share (CFS)

- CPU time is divided into time slices (epoch)
- some share of each epoch is given to the processes which are in ready queue.
- share is given to the process on the basis of their priority
- priority of every process is decided by its nice value
- nice values range ---> -20 to +19 (40 values)
  - \* -20 - highest priority
  - \* +19 - lowest priority

| Process | Nice Value |
|---------|------------|
| P1      | 10         |
| P2      | 10         |
| P3      | 10         |
| P4      | 10         |

Epoch - 100

| Process | Nice Value |
|---------|------------|
| P1      | 5          |
| P2      | 5          |
| P3      | 10         |
| P4      | 10         |



File : collection of data

File : data + meta data

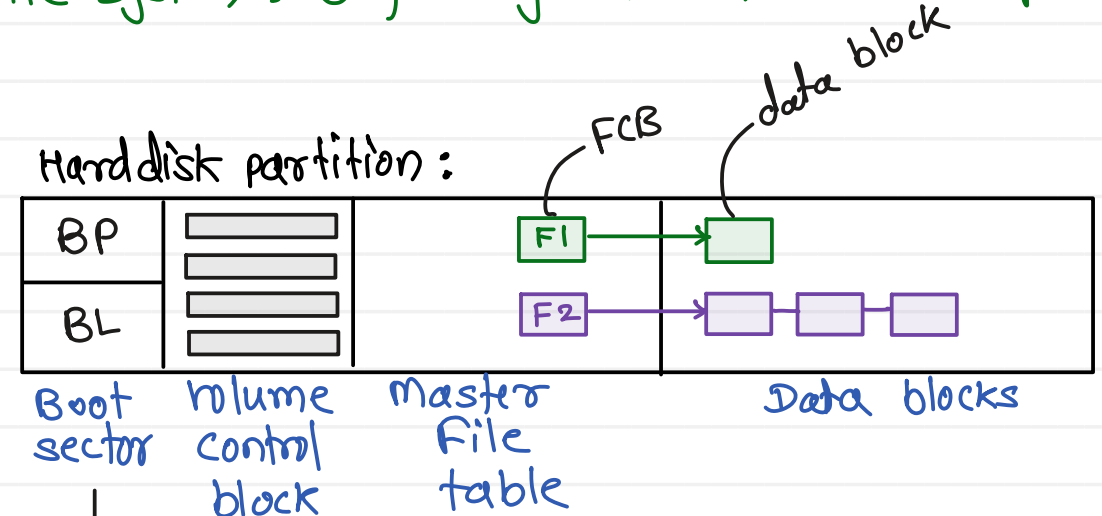
(Actual content of the file) (information about the file)

(data block) (File Control Block)

1. name, path
2. size
3. permissions  
r-read, w-write, x-execute  
levels - owner, group, other
4. owner, group
5. time stamps  
create/access/modify
6. type
7. information about data blocks

- size of data blocks is fixed but it can be configured at the time of formatting

File system : Organizing files on harddisk partition



programs required for booting

1. Bootstrap program
2. Boot loader

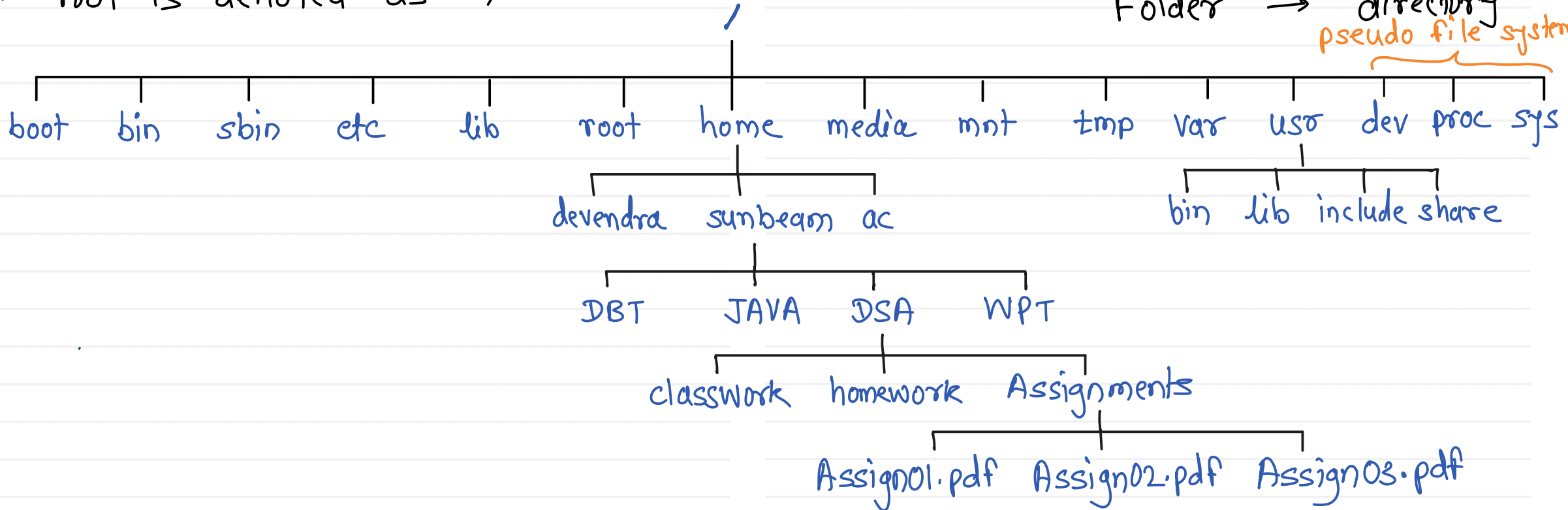
info about volume

- label, size
- empty, filled
- list of free data blocks

# Linux file structure

- Linux follows root file structure, which starts at root.
- root is denoted as '/'

Administrator → super user  
admin → root  
file → file  
Folder → directory  
*pseudo file system*



## 1. Absolute path :

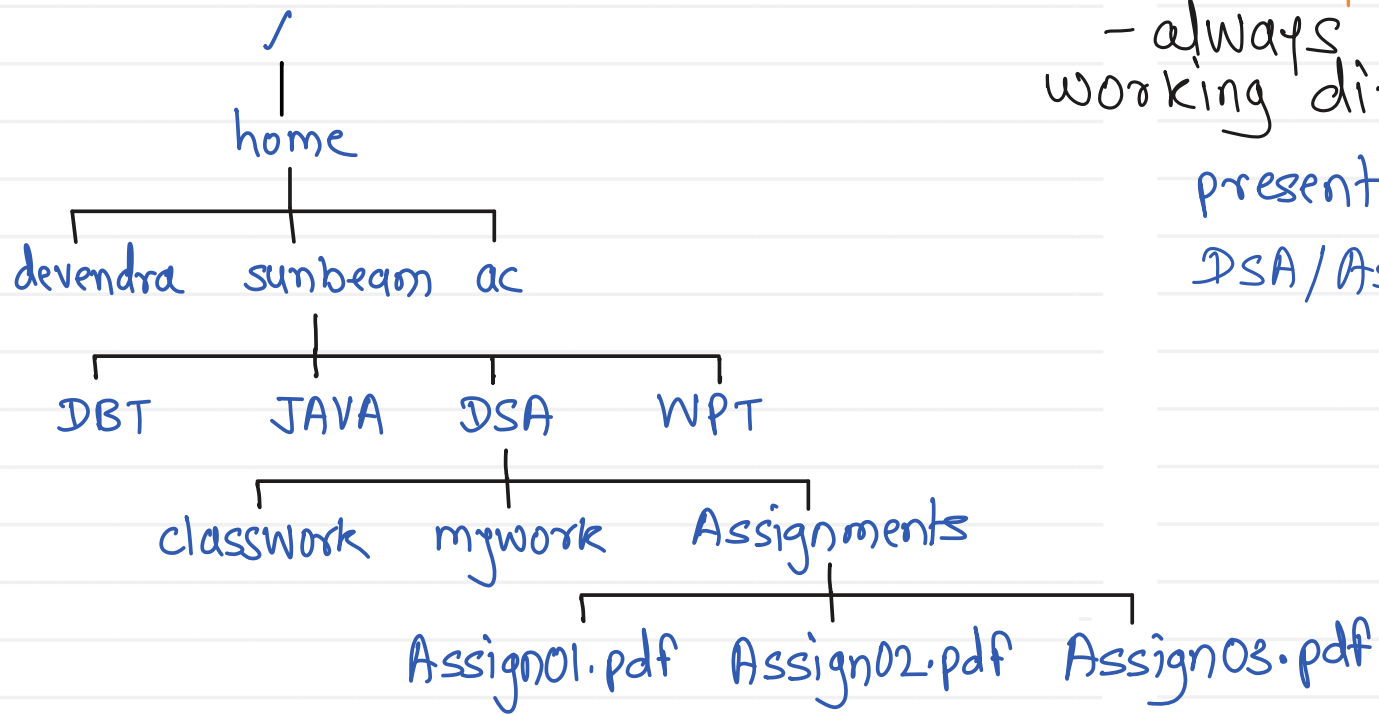
- always starts with /

/home/sunbeam/DSA/Assignments/Assign02.pdf

## 2. Relative path :

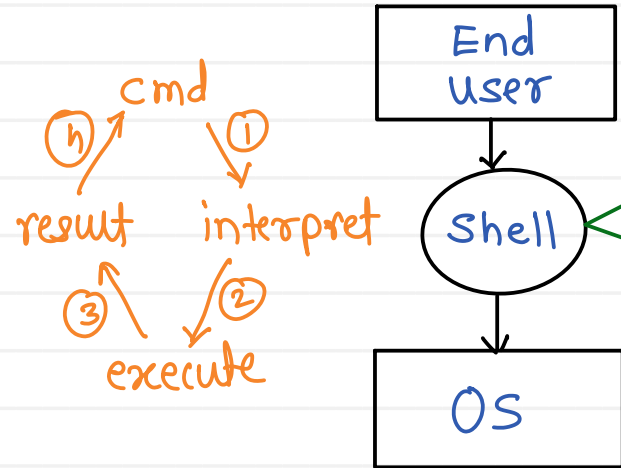
- always starts wot current/present working directory.

present working directory : /home/sunbeam  
DSA/Assignments/Assign02.pdf



- user interfacing is provided by implementing one program which is called as "shell"
- shell is intermediate between end user & operating system
- shell is a command interpreter

- in Vbuntu default shell is 'bash'  
BASH - Bourne Again Shell  
echo \$SHELL - will show default shell  
chsh - to change default shell



## CLI based shell

Windows : cmd.exe, powershell.exe

Linux : bsh(sh), bash, csh, ksh, zsh...

## GUI based shell

Windows : explorer.exe

Linux :

- GNOME (GNU Network Object Model Environment)
- KDE (Kommmon Desktop Environment)